# Equivalent Modeling of Cloud Data Center Overlay Digital Twins Based on Instance Deduplication

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***Abstract:*** With the rapid evolution of Intent-Driven Networks (IBN) powered by Large Language Models (LLMs) in cloud data centers (DCs), digital twins have emerged as a critical sandbox to verify network intents and prevent LLM configuration hallucinations. However, traditional 1:1 full-scale simulation based on Underlay physical topologies suffers from severe resource explosion and verification latency. This is primarily due to the massive multiplication of physical instances introduced by distributed anycast gateways (EVPN-VXLAN). To achieve second-level closed-loop verification, this paper proposes a lightweight digital twin modeling method oriented toward the Overlay logical layer.

Capitalizing on the transparency of VXLAN protocols to upper-layer services, this approach strips away underlying physical details and performs "logical deduplication" on Virtual Routing and Forwarding (VRF) and Bridge Domain (BD) instances scatteredly deployed across multiple physical Leaf switches. We construct a mathematical model for simulation overhead under equivalent service verification constraints, theoretically demonstrating that the state space complexity is reduced from $\mathcal{O}(N^2 + T \cdot N)$ to $\mathcal{O}(T)$, where N denotes the physical network scale and T denotes the total tenant scale.

Multi-dimensional Monte Carlo simulations show that while ensuring complete equivalence in connectivity and isolation verification, the proposed method achieves up to a 61.78-fold resource space compression in a large-scale network scenario featuring 100 Leaf switches and 300 tenants. Consequently, this renders the twin state space fully immune to physical node expansion with an O(1) constant property. This lightweight infrastructure significantly improves the verification efficiency of digital twins, providing an ultra-fast and low-overhead anti-hallucination verification foundation for future LLM-driven IBN toward second-level autonomous closed-loop operation.

***Key words:*** Digital Twin; Intent-Driven Network (IBN); Cloud Data Center; Overlay Abstraction; Logical Instance Deduplication

## 1. Introduction

In recent years, the deep integration of Intent-Driven Networking (IBN) and Large Language Models (LLMs) has emerged as a core paradigm shift for next-generation intelligent network operations (NetOps)[1]. Leveraging their powerful natural language understanding capabilities, LLMs can directly translate abstract service intents into concrete network configuration scripts. However, generative large models suffer from inherent "hallucinations," making their generated configuration instructions highly susceptible to high-risk anomalies such as routing conflicts and isolation failures. Therefore, before network configurations are officially deployed to the production network, pre-compliance verification within a digital twin simulation environment is strictly required.

In modern cloud data center scenarios, network architectures universally adopt a Spine-Leaf topology based on distributed anycast gateways, such as EVPN-VXLAN[2]. To support the large Layer 2 roaming of virtual machines (VMs) across arbitrary physical servers, instances of the same tenant's

Virtual Private Cloud (VPC), Virtual Routing and Forwarding (VRF) domains, and Layer 2 Bridge Domains (BDs) are highly scattered and instantiated across multiple physical Leaf switches. In traditional 1:1 Underlay full-scale physical digital twin modeling, to verify configuration correctness, the system must launch corresponding network operating system (NOS) images or processes for every single physical device. Furthermore, it must fully simulate the full-mesh VXLAN tunnels across the underlying network[3].

This traditional full-scale simulation suffers from a severe "physical replica explosion" problem. For instance, consider a typical tenant (VPC) in a cloud data center designed with 1 logical VRF and 2 logical BD subnets. Suppose BD1 connects 4 VMs and BD2 connects 3 VMs. Due to the randomness of cloud computing scheduling, these 7 VMs happen to be scattered across 7 different physical Leaf switches. If a 1:1 physical full-scale simulation is adopted, the system must run identical VRF processes, BD instances, and IRB Layer 3 interface configurations individually on 7 distinct virtual switch instances. When the entire network contains T tenants and the VMs of each tenant are highly dispersed, the number of physical configuration instances inside the twin sandbox will explode exponentially with the product of the number of tenants and devices. Consequently, constructing such a digital twin demands massive hardware computing resources, and the verification cycle is prolonged to the minute level. This thoroughly negates the practical value of using digital twins to safeguard second-level closed-loop intent execution in LLM-driven IBN.

To address this challenge, this paper proposes a lightweight digital twin modeling method oriented toward the Overlay logical layer. The core contributions of this paper are summarized as follows:

① We propose a lightweight Overlay digital twin architecture for cloud data centers based on logical instance deduplication that strips away the underlying Underlay physical dependencies. This architecture completely decouples the simulation overhead from the scale of physical switches.

② We establish a state space mathematical model under equivalent service verification constraints, rigorously proving that the proposed method reduces the complexity from $\mathcal{O}(N^2 + T \cdot N)$ to $\mathcal{O}(T)$.

③ Through multi-dimensional Monte Carlo randomized simulations, we quantitatively confirm that the proposed method achieves over a hundredfold compression of the resource state space in large-scale network environments, demonstrating complete immunity to physical topology expansion.

# 2. Architecture Mapping and Service Equivalence

## 2.1 Lightweight Overlay Mapping Architecture

The lightweight modeling principle proposed in this paper is summarized as "abstracting the underlying layer, deduplicating the higher layer, and ensuring service equivalence." This method breaks away from the traditional reverse modeling approach of "constructing physical devices before deploying service configurations." Instead, it directly builds logical twin entities based on higher-layer service entities, including VPCs, VRFs, and BDs.

The specific mapping principles are defined as follows: A tenant's Virtual Private Cloud (VPC) is abstracted into a globally unique, deduplicated logical node of a Layer 3 routing isolation domain (logical VRF). Subnets within the VPC are abstracted into deduplicated logical nodes of Layer 2 broadcast domains (logical BDs). Virtual Machines (VMs), serving as terminal nodes, are completely decoupled from their physical host tags and directly attached to their corresponding logical BD nodes.

In this lightweight Overlay twin model, physical Leaf switch entities and the full-mesh physical VXLAN tunnels interconnecting them are entirely eliminated. The mapping metadata, which tracks how VRF and BD instances are scatteredly deployed across specific physical Leaf switches in the production network, is maintained within an in-memory mapping table by the SDN controller and is not deployed as independently running entities inside the digital twin sandbox [4].

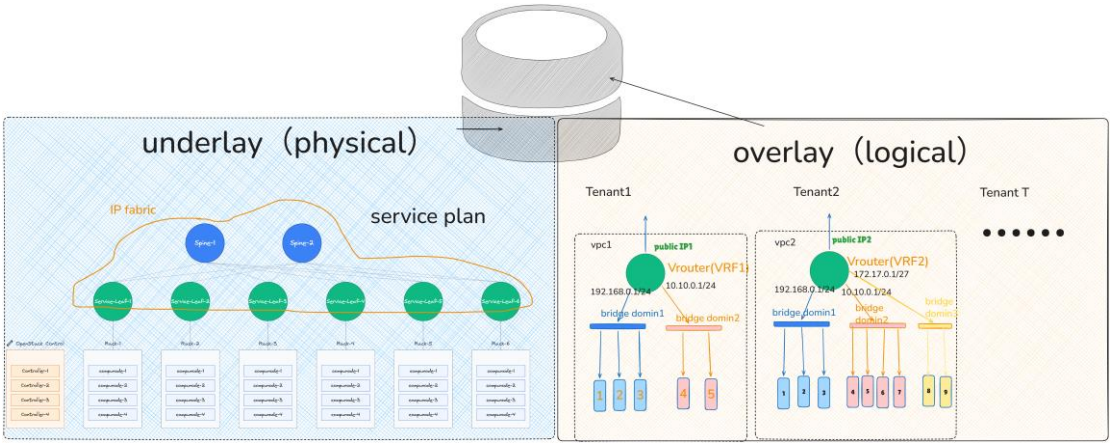



**Fig.1.** Architectural comparison and conceptual schematic between the 1:1 Underlay full-fidelity physical modeling and the Overlay logical deduplication modeling in cloud data centers.

## 2.2 Strict Definition of Service Equivalence and Justification

To guarantee that the lightweight model can accurately intercept configuration hallucinations generated by LLMs after discarding underlying physical details, we provide a mathematically rigorous definition of "service equivalence."

Let $\boxtimes$denote the set of intent-driven network security policy verification tasks generated by the LLM (including checking inter-tenant VPC hard isolation, subnet internal connectivity, cross-subnet routing reachability, and ACL access control policy compliance). Let $M_u$ represent the traditional full-scale physical digital twin model, with its corresponding intent decision function denoted as $f_u(r), r \in \boxtimes$. Let $M_o$represent the deduplicated logical digital twin model proposed in this paper, with its decision function denoted as $f_o(r)$. The two models are defined as service equivalent if and only if their decision results for any arbitrary task are completely identical:

$$\mathrm{Equiv}(M_u, M_o) \Leftrightarrow \forall r \in \boxtimes : f_u(r) = f_o(r)$$

The core underlying assumption of this model is the "neglection of Underlay link congestion and underlying encapsulation details." This assumption is firmly justified by the current operational realities of industrial cloud data centers. Modern data centers universally employ highly redundant two-tier or three-tier CLOS topologies and run Equal-Cost Multi-Path (ECMP) routing to achieve uniform load balancing across the entire network fabric[5].

Empirical measurement data from production data centers indicate that the utilization of underlying links generally remains within a low-load, safe zone; the average utilization of edge links is typically below 10%, while the median utilization of aggregation links ranges from approximately 10% to 20%[6]. Since the VXLAN encapsulation mechanism itself is entirely transparent to upper-layer service data streams, underlying routing perturbations and physical encapsulations do not alter the logical correctness of upper-layer service configurations (such as misconfigured IPs or missing ACLs).

Therefore, stripping away these underlying physical details does not cause any degradation in the detection accuracy of upper-layer service connectivity and isolation verifications[7].

# 3. Mathematical Model for Simulation Overhead

To provide a rigorous theoretical foundation explaining the root cause of resource preservation, this section establishes asymptotic complexity state space models for both twin paradigms. Assume the network consists of $N$ physical Leaf (VTEP) nodes, and the current total number of tenants (VPCs) in the production network is $T$. For the i-th tenant（$1 \le i \le T$）, the number of planned logical BDs is denoted as $b_i$.

## 3.1 Overhead of Traditional Underlay 1:1 Full-Fidelity Model($M_u$)

In full-scale physical simulation, the digital twin system must construct individual replicas for each tenant's gateway instances on every physical Leaf switch where they are scatteredly deployed in reality[8]. In the worst-case scenario where tenant applications are maximally dispersed due to live migration, let $S_{bd}(i,j)$ denote the set of physical Leaf switches where the j-th BD of the i-th tenant is actually deployed, and let $S_{vrf}(i)$ denote the set of physical Leaf switches hosting the corresponding Layer 3 VRF. The total runtime state space resource overhead $S_u$ for the full-scale model $M_u$ can be formulated as follows:

$$S_u = \sum_{i=1}^{T}\left(\sum_{j=1}^{b_i} |S_{bd}(i,j)|\right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}$$

Where $|\cdot|$ represents the cardinality of the set (i.e., the absolute number of physical Leaf switches across which that specific service instance is scattered); $C_{leaf}$ denotes the baseline constant overhead of running a single physical switch's native NOS operating system; $\frac{N(N-1)}{2}$ represents the total number of full-mesh VXLAN tunnels generated by the pairwise interconnection of $N$ physical nodes; and $C_{tun}$ represents the constant overhead required to maintain the dynamic protocol states of each physical tunnel within the twin data plane.

According to the operational characteristics of distributed anycast gateways, as the physical network scale $N$ and the tenant scale $T$ expand simultaneously, the physical deployment terms $|S_{bd}|$ asymptotically approach $N$， This causes the number of physical configuration replicas in the first two terms to explode at a composite scale of $\mathcal{O}(T \cdot N)$ Concurrently, the final term representing physical tunnel overhead exhibits a rigid quadratic growth of $\mathcal{O}(N^2)$ due to its full-mesh nature. Therefore, the upper bound of the overall state space complexity for the full-scale model $M_u$ is expressed as:

$$S_u = \mathcal{O}(N^2 + T \cdot N)$$

## 3.2 Overhead of the Proposed Overlay Logical Deduplication Model($M_o$)

In the Overlay twin model $M_o$, all physical nodes and full-mesh tunnels are completely decoupled from the underlying layer. Furthermore, global service deduplication is performed on identical configurations across different physical switches. Consequently, no matter how widely a BD or VRF is distributed across the production network, it is instantiated exactly once within the twin sandbox. Its total state space overhead $S_o$ is defined as:

$$S_o = \sum_{i=1}^{T} (b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}$$

Where $C_{obj}$ represents the unit constant overhead of a single pure logical service object (VRF or BD) on the deduplicated infrastructure; $E_{logic}$ denotes the total number of basic topological logical links between VMs and their corresponding logical BDs within tenants; and $C_{link}$ represents the constant maintenance overhead of a single logical link.

In actual cloud data centers, the number of subnets $b_i$ and the number of VMs defined by a single tenant are governed solely by the tenant's own application architecture. These parameters remain finite small constants that do not scale with the physical device count $N$ (typically bounded between 2 and 6)[9]. Consequently, the average logical overhead per tenant can be consolidated into a constant $K$, simplifying the equation to:

$$S_o = \sum_{i=1}^{T} K = K \cdot T$$

This theoretically proves that the space complexity of the deduplicated logical model $M_o$ scales strictly linearly with the tenant capacity, completely decoupling from the number of physical switches $N$:

$$S_o = \mathcal{O}(T)$$

## 3.3 Derivation of Theoretical Compression Ratio

To intuitively evaluate the reduction efficiency of the proposed lightweight method on the twin state space, we define the theoretical Compression Ratio $CR$ as the ratio of the traditional full-scale model overhead $S_u$ to the proposed overlay logical deduplication model overhead $S_o$:

$$CR = \frac{S_u}{S_o} = \frac{\sum_{i=1}^{T} \left( \sum_{j=1}^{b_i} |S_{bd}(i,j)| \right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}}{\sum_{i=1}^{T} (b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}}$$

When the physical node scale $N$ is small and the tenant scale $T$ dominates, the logical link terms in the denominator and the physical constants in the numerator are overwhelmed by the scale of growth. To isolate and evaluate the core service instance compression efficiency yielded by distributed anycast gateway deduplication, the above equation can be asymptotically approximated as:

$$CR \approx \frac{\sum_{i=1}^{T} \left( \sum_{j=1}^{b_i} |S_{bd}(i,j)| + |S_{vrf}(i)| \right)}{\sum_{i=1}^{T} (b_i + 1) \cdot C_{obj}}$$

This formula mathematically reveals the fundamental logic behind the resource savings: the numerator represents the summation of redundant service instances forced onto physical devices by distributed gateways due to VM dispersion; the denominator represents the total number of pure, unique logical service objects after deduplication. Regardless of how the tenant scale $T$ varies, since the numerator and denominator remain of the same order, the compression ratio $CR$ stabilizes within a constant range determined by the physical dispersion breadth of the production network's distributed gateways (i.e., the average number of Leaf switches across which each tenant is scattered, $|S_{bd}|$. Conversely, when the physical network scale $N$ (Leaf count) undergoes massive horizontal expansion, the $\frac{1}{2} N^2 \cdot C_{tun}$ term in the numerator dominantly takes over quadratically, causing the $CR$ to skyrocket by orders of magnitude as $N$ increases.

# 4. Simulation Evaluation and Result Analysis

## 4.1 Simulation Setup and Key Metrics

To quantitatively compare the evolutionary trends of both models, a large-sample Monte Carlo random simulation evaluation system was implemented (the relevant evaluation algorithm framework and open-source replication scripts are provided in the open-source link at the end of this paper). The baseline simulation parameters were set as follows: the physical network scale ranged within $N \in [10,100]$ (covering scenarios from a single-POD architecture up to the upper limit of ultra-large-scale cloud campus networks); the tenant scale $T$ was scanned from 10 to 400[10]. The number of logical BDs for each tenant followed a uniform random distribution reflecting a typical three-tier architecture, i.e., $b_i \sim U(2,6)$, which simulates the highly disordered scattering of tenant assets across the live network[11]. The relative weight coefficients of different model entities were configured as constants based on the complexity of the underlying framework.

By executing simulation experiments with multi-dimensional independent variables, the operational outcomes of different cross-sections were integrated into a single data chart, from which key physical scale cross-section data were extracted. The final resource control performance is illustrated in Figure 2, with the key cross-sectional data detailed in Table 1.

**Table 1 Cross-sectional experimental data under different network scales N**

| Switch Count N | Full Model $S_u$ (T=50) | Deduplicated Model $S_o$(T=50) | Full Model $S_u$(T=300) | Deduplicated Model $S_o$ (T=300) | Compression RatioCR (T=300) |
|---|---|---|---|---|---|
| 10 | 1879 | 348 | 9415 | 2139 | 4.40 |
| 20 | 4860 | 328 | 22122 | 2127 | 10.40 |
| 30 | 7195 | 350 | 33095 | 2087 | 15.86 |
| 40 | 10964 | 360 | 45208 | 2078 | 21.76 |
| 50 | 15295 | 346 | 58747 | 2113 | 27.80 |
| 60 | 20286 | 368 | 72156 | 2075 | 34.77 |
| 70 | 25063 | 342 | 84535 | 2093 | 40.39 |
| 80 | 30264 | 347 | 98584 | 2078 | 47.44 |
| 90 | 35991 | 348 | 111693 | 2070 | 53.96 |
| 100 | 43110 | 365 | 129560 | 2097 | 61.78 |

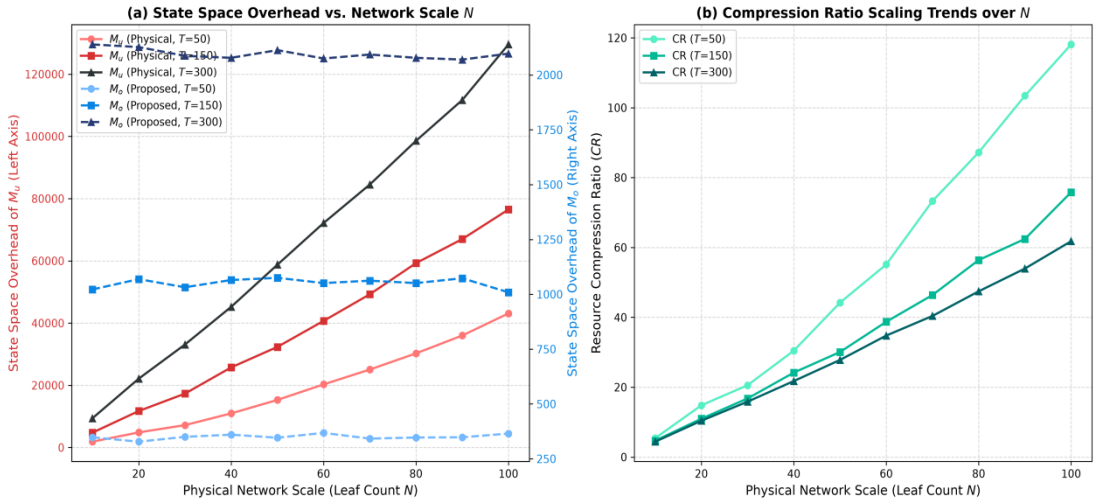



**Fig.2 Comparison curves of twin state space resource overheads under multi-dimensional variable evolution**

## 4.2 Interpretation of Experimental Conclusions

By comparing the cross-sectional data in Table 1 with the dynamic curves in Figure 2, two core academic conclusions can be drawn:

①Complete immunity to physical topology expansion ($\mathcal{O}(1)$ property): the $S_o(T = 300)$ column in Table 1, when the underlying network scale N of physical switches surges multi-fold from 10 to an ultra-large cluster of 100, the overhead of the traditional full-scale model $S_u$ grows quadratically due to the joint impact of $\mathcal{O}(N^2)$ physical tunnel states and distributed anycast replica explosions, eventually reaching a state space overhead of 129,560. In sharp contrast, the state space overhead of the proposed Overlay logically deduplicated twin model consistently remains within a narrow corridor around 2,100 (subtly fluctuating from 2,139 to 2,097), which manifests as a perfectly horizontal constant line in Figure 2. This absolute immunity of overhead against physical network horizontal scaling quantitatively validates that the deduplication algorithm successfully eliminates all quadratic and multiplicative terms containing N, thereby empowering the digital twin sandbox with a "dimension-reduction immunity" attribute from a high-level business perspective.

② The larger the physical network scale, the more the deduplication compression gains leap by orders of magnitude: As anticipated by the mathematical model derivation in Chapter 3, with the scaling of the data center's physical topology size N, redundant entries forced onto physical switches by distributed gateways accumulate geometrically. When fixing the tenant scale at T=300, the resource compression ratio (CR) exhibits a steady linear upward trend as the number of physical nodes N increases. In a small-scale private cloud single-POD scenario where N=10, the compression ratio is a modest 4.40-fold; however, when the network horizontally scales to an ultra-large public cloud campus scenario where N=100, the resource compression multiplier scales up to 61.78-fold. This demonstrates that the efficiency of the proposed algorithm does not degrade with the expansion of the network scale. Instead, it exhibits excellent scalability characterized by "the larger the physical network, the more prominent the asset savings," which serves as the efficiency cornerstone for deploying pre-verification sandboxes into large-scale DCs to safeguard the second-level safety closed-loops of LLM-driven IBNs.

# 5. Conclusion and Future Work

Aiming at the state space explosion problem of digital twins caused by distributed anycast gateways in cloud data centers, this paper proposes a lightweight digital twin modeling method

oriented toward the Overlay logical layer. By establishing a state space mathematical model under equivalent verification constraints, we theoretically prove that the new scheme eliminates all quadratic and multiplicative terms containing the physical node count N by stripping away Underlay physical details and performing high-level logical deduplication, successfully reducing the state space complexity from $\mathcal{O}(\mathrm{N}^2 + \mathrm{T} \cdot \mathrm{N})$ to $\mathcal{O}(\mathrm{T})$ .Multi-dimensional Monte Carlo random simulation experiments show that while ensuring complete equivalence in connectivity and isolation verification, the proposed deduplicated logical model possesses complete O(1)-level constant immunity against the horizontal expansion of the underlying physical network. In a large-scale network scenario featuring N=100 physical switches and a tenant scale of T=300, this method achieves up to a 61.78-fold compression of the resource space. This approach provides a core theoretical foundation and quantitative experimental support for the lightweight deployment of digital twins in ultra-large-scale data centers.

The ultimate value of the proposed method lies in its role as a millisecond-level pre-deployment safety sandbox validation engine before LLM configurations are pushed to production. In future intent-driven networks, network configuration deployment will undergo a three-stage closed-loop validation process. In the first stage (Intent Translation and Pre-deduction), after a user inputs a natural language business intent, the LLM translates it and generates specific underlying configuration scripts. In the second stage (Lightweight Sandbox Millisecond-level Safety Check), the scripts are fed into the deduplicated Overlay digital twin proposed in this paper. Since this model strips away all redundant physical entities and tunnel encapsulations, the vertices and edges of the graph topology are substantially compressed. Consequently, running routing reachability traversals or access control list (ACL) rule conflict retrievals within the twin environment drops its computational latency from minutes in traditional physical simulations to milliseconds. The sandbox can instantly determine whether the LLM has generated a configuration hallucination and execute either an approval or an interception. In the third stage (Physical Deployment), the logic configurations verified to be hallucination-free are handed over to the authoritative SDN controller, which completes the final distributed 1:N replica instantiation expansion and configuration deployment onto the live physical switches. This closed-loop mechanism, which completely decouples high-level logical validity verification from low-level physical distributed deployment, effectively bridges the native gap between the second-level safety enforcement requirements of large models and the high latency of traditional physical network simulations.

In future research, primary efforts will be dedicated to advancing the intermediate translation and mapping technologies between the digital twin network and the live network, automatically translating the full Underlay network (physical switches, distributed gateways, and full-mesh tunnels) into an Overlay logical digital twin network, thereby completing the automated mapping pipeline between live configurations and the twin model. Through configuration collection, normalization, and deduplication translation from production environments, the lightweight twin can be instantaneously generated based on the actual live topology and incrementally synchronized upon configuration changes; simultaneously, the logic configurations verified in the sandbox will be reversely translated into live physical configurations to interface with the physical deployment phase of the three-stage closed loop, ultimately enabling the millisecond-level pre-deployment check mechanism proposed herein to truly land in large-scale production networks.

**Data and Code Availability Statement: The full source code of the Monte Carlo random simulation evaluation system, experimental simulation data, and high-definition academic charts involved in this study are fully open-sourced and hosted on the GitHub repository. Readers and peers can access and execute complete replication via the following link:*https://github.com/qiuqiueyu/Overlay-DigitalTwin-Deduplication***

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